Saturated mixed block systems at a crossing

Human Compression project: bounded L044 proof component

Retain the unique-crossing setting in $G \le A_n$, $n \ge 12$, with all maximal omissions intransitive. The two G_i-orbits are A, B, with crossing endpoints a, b. The deficits and minimal simultaneous block-image family M are those of the accepted block-deficit note. Block systems are chosen maximal, so their induced actions are primitive. The arguments below assume $\alpha = \beta = 0$ unless stated otherwise.

Lemma 1 (A binary and a maximal symmetric block image)**.** *Suppose the block numbers are $2, m$, with $m \ge 3$. The actual M-action is faithfully the diagonal $C_2 \times S_m$ action. Its distinguished C_2 generator and its pure S_m component occur on opposite sides of i. The entire left and right intervals of G_i are supported on their respective original orbits, after reversal if necessary.*

Proof. Saturation gives the full block image $C_2 \times S_m$ and makes its lift faithful, by the paid actual-orbit argument. Choose a minimal subfamily $T \subset M$ generating the S_m quotient. Normal-quotient normalization bounds the remaining independent family by one. Whiston then forces $|T| = m - 1$, leaving exactly one distinguished generator e outside T.

The maximal involutory string images of T have a connected noncommuting path. This follows from the accepted extremal classification for $m \ge 5$, and from the complete elementary S_3, S_4 cases in the anchored symmetric lemma. Their indices therefore form an interval. Since their image is transitive, one of them moves the block containing b, so that interval contains $i - 1$ or $i + 1$; reverse so it starts at $i + 1$. Only its first member moves that block.

The S_m image of e is trivial. If e lies on the opposite side of i, a nontrivial image would disconnect the active set of this primitive block action, contrary to the paid primitive interval lemma. If it lies on the same side, it is beyond T, fixes the crossing block, and centralizes its far-end omission (or all of T). The anchored symmetric lemma makes its image trivial in this case too. Thus e is the pure nonidentity C_2 element. Its index must be $i - 1$, since it exchanges the two blocks containing a and cannot lie beyond T.

The subgroup generated by T intersects this central C_2 trivially, by IP in the faithful block action. It is a character graph $S_m \to C_2$. That character is trivial: the $m-2$ tail generators of T fix the crossing block and are independent in S_{m-1}, so generate it by Whiston's equality clause. Some has odd S_m image. All have indices at least $i + 2$ and fix a, so have trivial C_2 image. A nontrivial character of S_m, namely its sign, is therefore impossible. The pure components are on opposite sides as claimed.

Every generator on the S_m side fixes a: its near generator is now pure on the other original orbit, and the more distant ones fix a by the crossing property. The accepted commuting anchor lemma makes this whole interval trivial on the orbit of a. The symmetric statement holds for the other interval and b. This proves exact support separation, not merely separation of the block images. □

Lemma 2 (The remaining saturated $D = 2$ case)**.** *The residual block systems $(k_A, m_A) = (4, 3)$ and $(k_B, m_B) = (k, 2)$, with $\alpha = \beta = \gamma = 0$, are impossible.*

Proof. Lemma 1 separates the two interval supports. On the twelve-point orbit, M is the natural S_3 on four copies, with two consecutive generators at the end next to i. Only its one outer neighbor can fail to centralize all of this S_3. Any further distinguished generator fixes the crossing endpoint and therefore fixes its whole three-point copy. The full centralizer on the

remaining three copies is S_3 permuting those copies. A transposition of three-point copies is odd on the twelve points. Since this interval fixes the other original orbit pointwise, its even part is A_3, which contains no nonidentity involution. No such further generator exists. This interval has rank at most three.

On the other orbit the last distinguished generator is its pure swap of two k-point blocks. Apart from its possible outer neighbor, every other generator centralizes that swap and fixes the crossing endpoint; it acts faithfully on the $k-1$ nonanchor row pairs. Whiston bounds their number by $k-2$, so this interval has rank at most k. This includes $k=2$. Thus the sum of the two interval ranks is at most $k+3$. But the exact deficit formula under the stated saturation gives $r=k+6$, hence that sum is $r-1=k+5$. This contradiction excludes the case. □

Lemma 3 (Saturated mixed two-block and two-point-block systems)**.** *For block systems* $(k,2)$ *and* $(2,m)$*,* $m\geq 3$*,* $\alpha=\beta=0$ *implies* $r\leq k+m-1=(n-2)/2$*.*

Proof. By Lemma 1 the original supports separate. On the two-point-block orbit the $m-1$ generators of the pure S_m component end next to i. A farther generator centralizing all of S_m fixes the crossing endpoint and its two-point block pointwise. Commutation transports this fixation to every block, so it is trivial. The only possible additional generator is the one outer neighbor x of the S_m interval. Its block image centralizes the far-end omission and fixes the crossing block, so is trivial by the anchored symmetric lemma. Pointwise fixation of the two-point crossing block then propagates to at least $m-1$ blocks. On at most one remaining two-point block x may be a transposition. Its action on the other original orbit is trivial by support separation, so evenness excludes that possibility too. Thus this interval has exactly $m-1$ generators.

On the two-block orbit, the pure swap has k transpositions, so k is even. Its outer neighbor must exist and must move the opposite anchor to another row. If $k=2$ this is one odd transposition, impossible. For $k\geq 4$ the proof of the accepted paired-row side-rank lemma applies without any assumption on the number of blocks in the opposite orbit. For clarity, its used hypotheses and endpoint step are as follows. The diagonal tail has rank at most $k-2$, so the entire interval has rank at most k. At equality, its tail is a maximal S_{k-1} string. For even $k\geq 6$ the ordinary endpoint omission makes its outer neighbor a single odd transposition. For $k=4$ the complete two endpoint possibilities are

$$(1\ 2)(0'\ 3') \quad \text{and} \quad (1'\ 2')(0'\ 3').$$

After deleting the far tail generator, the remaining near tail, this endpoint, and the block swap are transitive on all eight points: their contracted row edges are 2–3, 0–3, and 1–2. The entire opposite interval is unchanged and transitive on its original orbit; ρ_i joins them. The global omission would be transitive, contradicting a standing hypothesis. Thus the two-block interval has rank at most $k-1$. Adding ρ_i gives $r\leq (k-1)+(m-1)+1=k+m-1$. □

Lemma 4 (The one-deficit mixed block image)**.** *For block systems* $(k,2)$ *and* $(2,m)$*,* $m\geq 3$*, the case* $\alpha=1$*,* $\beta=\gamma=0$ *cannot violate the rank bound.*

Proof. Suppose it does. The exact deficit gives $r=k+m$, $|M|=m-1$, and k distinguished generators outside M and i. The zero orbit deficit makes the block action faithful on M, with diagonal natural actions on k binary rows and two m-point copies. Let H be its primitive degree-m block image. Its active indices form one interval next to i by the accepted primitive interval lemma; reverse so that this interval starts at $i+1$.

First suppose there is a member of M outside that interval. Its image on H is trivial, so minimality makes it the sole such member and the pure C_2 generator ρ_{i-1}. The remaining $m-2$ generators project independently onto H: any projected redundancy could be canceled by this available pure binary generator. IP makes their subgroup a character graph over H in $C_2\times H$. That graph is trivial on the binary coordinate. Otherwise only its first generator could have

nontrivial binary image, since all its later generators fix the crossing endpoint. Both neighbors of i would then swap the paired anchor in A, leaving that pair invariant under G_i by the same commutation argument used in the two-swap proof. This is impossible. The commuting anchor lemma now separates the full left and right supports.

The two-block side has rank at most $k-1$ by the complete side-rank argument in Lemma 3 (including the exclusion at $k=2$). The other side consists of its $m-2$ members of M and at most one outer neighbor. To justify the latter assertion, a farther generator centralizes all of H and fixes the crossing endpoint. It therefore fixes the entire natural copy containing that endpoint, and preserves the other copy. An involution centralizing a primitive group of degree at least three is the identity: if nontrivial it is fixed-point-free, and its two-point orbits form an invariant nontrivial block system. Thus the farther generator is also trivial on the second copy. Support separation makes it trivial everywhere. The total rank is consequently at most $(k-1)+(m-1)+1 = k+m-1$, a contradiction.

It remains that all of M is the single interval starting at $i+1$. Only its first generator has nontrivial binary image, and only that generator can move the degree-m crossing block. The accepted binary anchored-string lemma applies: the H image is S_m and its far-end omission has an orbit containing the crossing block of size at least $m-1$, with centralizer fixing that block trivial. For $m=3$, not stated in that lemma, the assertion is immediate: the last generator is the transposition of the other two blocks, the first projects to a transposition moving the crossing block, and its one-generator far-end omission has the required two-point orbit and centralizer property.

Let $u=\rho_{i-1}$ and $v=\rho_{i+m}$ when those indices exist, and let C be the other generators outside M and i. They centralize M and fix both crossing endpoints. On the two m-point copies their action is trivial: fixing one point fixes the first copy, while the natural S_m has trivial centralizer on the second. On A they are diagonal and fix its anchor row. Hence their action is faithful in S_{k-1} and they number at most $k-2$. There are k outside generators, so both u, v exist and C has exactly $k-2$ members. Whiston's equality clause gives $\langle C\rangle = S_{k-1}$ on the nonanchor rows (trivial if $k=2$).

The generator v fixes both crossing endpoints and centralizes the far-end omission of M. Its degree-m block image is therefore trivial. It fixes the two-point crossing block pointwise, and commutation propagates this fixation to at least $m-1$ blocks. On the remaining orbit it can thus move at most the final two-point block, by one transposition. On A, the generator v commutes with the first member of M (their index distance is $m-1 \geq 2$) and preserves its two blocks. It acts diagonally and hence evenly there. Evenness excludes a single transposition on the other original orbit. Thus v is trivial on that whole orbit. Its remaining diagonal action on A fixes the anchor row and lies in the already full $\langle C\rangle = S_{k-1}$. The distinguished generator v is redundant, the final contradiction. □

Boundary. These proofs retain the full accepted L047 independent-generation and anchored symmetric interfaces, the L044 block-deficit proof, and the two-block completion. They do not assert that a block quotient has IP unless faithfulness has been obtained from $\beta = 0$. Only the mixed $\alpha = 0, \beta = 1, \gamma = 0$ case remains separate in this numerical residual family, as do the other outstanding principal source obligations. Source framework: Cameron–Fernandes–Leemans–Mixer, arXiv:1605.09173v1, Section 5.1. No new census, Lean theorem, original or question credit is claimed.